

HAT-P-36b

R-1440

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_210520 · created 2026-08-01T08:18:20+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459354.7542 +/- 0.0038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.103 +/- 0.026
Transit depth (Rp/Rs)^2	1.07 +/- 0.53 [%]
Orbital Inclination (inc)	89.57 +/- 2.46
Airmass coefficient 1 (a1)	0.985 +/- 0.016
Airmass coefficient 2 (a2)	0.012 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.63 %
Best Comparison Star	#3 - [388, 204]
Optimal Aperture	3.74
Optimal Annulus	8.88
Transit Duration (day)	0.0943 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.103 ± 0.026 vs 0.126 (deviation 0.88σ)
Duration fit vs published	0.0943 d vs 0.0925 d (deviation 0.55σ)
Mid-time O–C	-4.1 min
Depth SNR	4.0
Residual scatter	1.63 %

Light curve

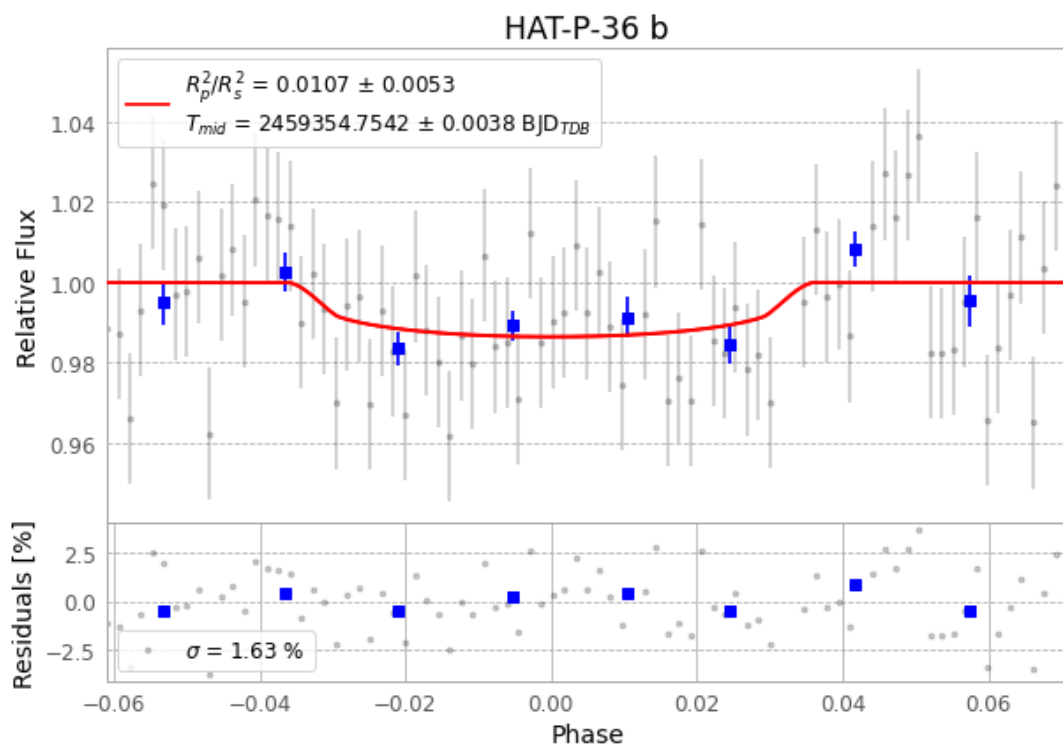


Figure 1 — FinalLightCurve_HAT-P-36 b_2021-05-19.png (SHA-256 4126c93ee81e7d78...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [395, 219], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b6c35986b484b9fe...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

83 FITS frames (55 MB), HATP-36210520040612.FITS → HATP-36210520081812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-210520.log (74242c109776...), FinalLightCurve_HAT-P-36 b_2021-05-19.png (4126c93ee81e...), FinalLightCurve_HAT-P-36 b_2021-05-19.csv (548be0873646...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 08:18Z.